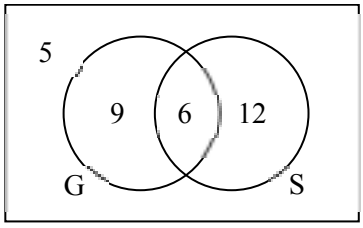
| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 7(d) | $a = \frac{1}{2}$ , $b = \frac{1}{2}$ | **5** | **B2** for 2 correct equations eg<br>$a + b = 1$, $8a + 4b = 6$<br>or **B1** for 1 correct equation<br><br>**B2** for one correct value<br>or **M1** (dep on at least B1) for correctly eliminating one variable from two linear equations in a and b<br><br>OR<br><br>**B2** for $a = \frac{1}{2}$<br>or **B1** for 6a = 3 or for 3rd difference = 3<br>**B2** for $b = \frac{1}{2}$<br>or **M1** for substituting *their* a into a correct equation of first differences |
| 8(a) | $ab(3a - b)$ final answer | **2** | **B1** for $a(3ab - b^2)$ or $b(3a^2 - ab)$<br>or $ab(3a - b)$ seen |
| 8(b) | $x > 7.5$ final answer | **2** | **B1** for $12+3 < 5x - 3x$ oe |
| 8(c) | $27x^6y^{12}$ | **2** | **B1** for two of 27, $x^6$ and $y^{12}$ correct |
| 8(d) | 0.5 or $\frac{1}{2}$ | **3** | **M2** for $4 = 6x + 2x$ or better<br><br>or<br>**M1** for $2(2 - x) = 6x$ oe |
| 8(e) | $2x^3 + 5x^2 - 23x + 10$ final answer | **3** | **B2** for correct expansion of three brackets unsimplified<br><br>**B1** for correct expansion of two brackets with at least 3 terms correct |
| 8(f)(i) | $200\left(1+\frac{r}{100}\right)^2 = 206.46$ oe | **M1** | |
| | $1 + \frac{2r}{100} + \frac{r^2}{100^2}$ oe | **M1** | |
| | $r^2 + 200r - 323 = 0$ | **A1** | Correct solution reached with no errors or omissions seen<br>If 0 scored, **SC1** for $200(n)^2 = 206.46$ |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 8(f)(ii) | $\dfrac{-200+\sqrt{200^2-4(1)(-323)}}{2\times1}$ | **B2** | **B1** for $\sqrt{200^2-4(1)(-323)}$ or $(r+100)^2$<br><br>**B1** for $\dfrac{-200+\sqrt{q}}{2\times1}$ or $r=\sqrt{323+100^2}-100$<br><br>OR<br>**B2** for $100\left(\sqrt{\dfrac{206.46}{200}}-1\right)$<br>or **B1** for $\sqrt{\dfrac{206.46}{200}}$ |
| | 1.60 cao final answer | **B1** | |
| | | | |
| 9(a)(i) |  | **2** | **B1** for two correct values<br><br>Or<br><br>**B1** 5 outside and total in G = 15 and total in S = 18 |
| 9(a)(ii) | $\dfrac{3}{8}$ oe | **1** | **FT** $\dfrac{\textit{their } 12}{32}$ |
| 9(a)(iii) | $\dfrac{2}{5}$ oe | **1** | **FT** $\dfrac{\textit{their } 6}{15}$ |
| 9(b) | 96 | **2** | **M1** for $\dfrac{36}{64}=\dfrac{54}{x}$ oe or $36=\dfrac{54}{(54+b)}\times100$ oe<br>If 0 scored **SC**1 for answer 150 |
| 9(c)(i) | $\dfrac{9}{25}$ oe | **2** | **M1** for $\dfrac{15}{25}\times\dfrac{15}{25}$ oe |
| 9(c)(ii) | $\dfrac{16}{25}$ oe | **1** | **FT** 1 – *their* **(c)(i)** |
| 9(d) | $\dfrac{17}{20}$ oe | **3** | **M2** for $1-\dfrac{10}{25}\times\dfrac{9}{24}$ oe<br>or for $\dfrac{15}{25}\times\dfrac{14}{24}+\dfrac{15}{25}\times\dfrac{8}{24}+\dfrac{15}{25}\times\dfrac{2}{24}+\dfrac{8}{25}\times\dfrac{15}{24}+\dfrac{2}{25}\times\dfrac{15}{24}$ oe<br><br>or **M1** for one correct relevant product |